

Video Emotion Recognition in DUUI

A Multimodal, Multi-Level Analysis Framework

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1 Issue of the Task

Emotion recognition in video streams has traditionally relied on isolated, unimodal analyses (e.g., analysing only facial frames). Human emotion—especially within complex, dynamic environments like political speeches or parliamentary debates—is inherently multimodal: a speaker may use neutral language while displaying aggressive facial action units and tense intonation, or smile while expressing verbal opposition.

The core challenge this project addresses is the implementation and evaluation of a robust, **multimodal** emotion recognition component within the DUUI framework.

1.0 Identified Baseline Problem

Initial testing with classic Vision Transformers (ViT-Base models trained on FER-2013 and AffectNet) revealed a severe **contextual head-pose bias**: both models produced approximately 75% “sad” classifications regardless of the speaker’s actual emotional state. Since both models share the same architecture (ViT-Base, Patch 16, 224×224 input) but differ only in training data, this effect is attributable to the video content itself—particularly head inclination—rather than any difference in training distribution. This confirms the inadequacy of purely frame-based facial analysis for real-world political video data.

2 Available Resources

2.1 Models

Model	trpakov/vit-face-expression	dima806/facial_emotions_image_detection
Architecture	ViT-Base (google/vit-base-patch16-224)	ViT-Base (google/vit-base-patch16-224)
Parameters	≈86M	≈86M
Disk Size	≈340 MB	≈340 MB
Training Data	FER-2013 (≈35k images, 48×48 grayscale)	AffectNet (≈440k annotated web images, colour)
Output Classes	7 (Ekman)	8 (Ekman + Contempt)

Both models serve as the **classic baseline**. They will be compared against multimodal LLMs (Phi-4, Qwen) which are expected to reason across modalities.

2.2 Emotion Taxonomies & Output Formats

- **Ekman classes**: Anger, Disgust, Fear, Happiness, Sadness, Surprise, Contempt
- **Valence / Arousal**: Continuous scales from -1 to $+1$
- **FACS (Facial Action Coding System)**: ≈40 Action Units (AUs) for micromimicry description

2.3 Technology Stack

DUUI, Python, HuggingFace Transformers, Phi-4, Qwen, REST APIs.

3 Requirements

1. **Multimodal Integration**: Ingest and synchronise video, audio, and text streams within DUUI.
2. **Architectural Comparison**: Formal benchmark of classic ViT models versus multimodal LLMs, focusing on temporal consistency and cross-modal conflict resolution.
3. **Multi-Level Temporal Analysis**: Emotion output must be structured across three levels (see Section 4).
4. **Divergence & Error Tracking**: Track sentiment-emotion divergence (e.g., neutral text with emotionally charged audio).

5. **Head-Pose Bias Mitigation:** The evaluation must specifically address and attempt to resolve the identified head-pose bias from baseline testing.

4 Multi-Level Analysis Framework

The system will produce emotion information across three temporal granularities:

Level	Unit	Description
Micro	Frame / Segment	Per-frame AU activation, Valence/Arousal, Ekman class. Captures micromimicry and rapid expression changes.
Meso	Speech / Speaker	Dynamics within a single speech: duration of emotional states, peaks (e.g., anger spikes), rapid shifts vs. stable phases, speaker-level emotional style.
Macro	Session / Day	Session-level progression: emotional escalation or de-escalation, contagion across speakers, influence of previous speeches on subsequent emotional baseline.

5 Research Questions & Analysis Dimensions

The following questions guide the evaluation and analysis work:

5.1 Emotion Consistency & Error Analysis

- Do video, audio, and text emotion signals agree, and which modality deviates most frequently?
- In which situations does each modality fail (text vs. audio vs. video)?
- How stable is the emotion signal over a sliding time window?

5.2 Sentiment vs. Emotion Divergence

- When is text neutral while audio is emotionally charged?
- When does video signal negative emotion while text carries positive content?
- Which modality is most reliable in a political speech context?

5.3 Cross-Modal Conflicts

- When do modalities contradict each other?
- Which conflicts lead to misclassification?
- Can an LLM resolve cross-modal conflicts better than a rule-based system?

5.4 AU-Based Analysis

- Which AUs correlate with specific emotions in a political context?
- Are there typical AU patterns for “controlled anger” or “neutrality”?
- How strongly do AU patterns differ between individual speakers?

5.5 Pose & Body Language

- Does open posture correlate with positive valence?
- What is the relationship between gestural intensity and audio-based emotion?

5.6 Stance / Attitude Detection

- Is a speaker supportive, opposing, or neutral toward the subject being discussed?
- Derived via: text (semantic), audio (intonation), video (non-verbal cues).

5.7 System Comparison

- Classic ViT vs. LLM (Phi-4 / Qwen): which architecture is more stable?
- Which system delivers better temporal consistency?

6 Use Cases

1. **Cross-Modal Conflict Detection in Political Contexts.** Detecting instances where modalities contradict—e.g., a politician displaying “controlled anger” (neutral text, aggressive AUs, tense intonation). The system will evaluate whether an LLM resolves this conflict better than a strict rule-based model.
2. **Speaker Profiling & Stance Detection.** Analysing a speaker’s baseline “emotional style” to determine their stance (supportive, opposing, neutral) based on consistency of valence/arousal and body language (e.g., open posture correlating with positive valence).
3. **Temporal Speech Dynamics & Escalation Tracking.** Monitoring a parliamentary session to detect emotional contagion—tracking whether a highly charged speech shifts the emotional baseline of subsequent speakers.

7 Implementation Plan

The project is divided into four sequential phases:

Phase 1 — Baseline Setup & Integration Integrate the two ViT-Base models into DUUI. Establish the data pipeline for frame extraction and map outputs to Ekman classes, Valence/Arousal, and AUs.

Phase 2 — Multimodal Pipeline Integrate audio processing and NLP text sentiment via REST. Implement the Micro, Meso, and Macro temporal tracking structures.

Phase 3 — LLM Integration Connect Phi-4 / Qwen as the multimodal reasoning engine, feeding combined modality signals to resolve cross-modal conflicts.

Phase 4 — Evaluation & Tuning Error analysis focused on head-pose bias. Benchmark LLM conflict resolution against classic models across all three analysis levels.

8 Project Schedule

